Yi Yang

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Professional Summary

Computer Science Ph.D. researcher developing molecular representation-learning methods and discrete 3D generative models using graph Transformers, fragment tokenization, and SE(3)-aware geometry. Published TMLR work on multi-scale molecular representations, with experience in reproducible PyTorch/PyG workflows, and pharmaceutical R&D.

Education

University of Texas at Dallas — Dallas, TX
Ph.D. in Computer Science, GPA: 3.8 — *Expected May. 2028*
- Erik Jonsson Graduate Scholarship in Computer Science, 2023

Research Experience

Graduate Research Assistant | University of Texas at Dallas — Aug. 2024 – Present

PoseWords: Mobility-Aware 3D Molecular Tokenization and Generation — *July 2026 – Present*
- Investigating molecular topology and geometry representations for peptide-conditioned ligand generation, studying how discrete encodings capture 2D connectivity and 3D conformations.
- Developed a mobility-aware tokenizer with fragment conformer quantization; tokenizer evaluation showed 653% greater fragment reuse, 41% fewer single-atom tokens, and 14% shorter sequences versus a rule-based baseline.
- Developing an SE(3)-aware generative framework with torsion-based geometry representations and chemically constrained fragment assembly; controlled evaluations and geometric error analysis guide ongoing representation redesign.

BiScale-GTR: Fragment-Aware Graph Transformer [Paper Link] — *Oct. 2025 – July 2026*
- Developed BiScale-GTR, a self-supervised, fragment-aware graph Transformer combining GINE atom encoders, attention pooling, gated atom-fragment fusion, and a structure-aware Transformer for multi-scale molecular representation learning.
- Built a deterministic RDKit graph-BPE pipeline with WL hashing, chemical-validity filtering, and recursive OOV decomposition; improved Peptides-func AP from 0.632 to 0.672 and Peptides-struct MAE from 0.293 to 0.262 using 78.5% less pretraining data than 2M-molecule baselines.

Interpretable ML for Organic Photovoltaic Property Prediction [Paper Link] — *Sep. 2024 – Aug. 2025*
- Built an end-to-end explainable XGBoost pipeline to curate molecular data, compute 2,000+ descriptors, train regression models, and predict photovoltaic performance across 350 donors and six acceptors, achieving test $R^2 = 0.918$.
- Performed feature selection, SHAP-based attribution and performance analysis, and external-dataset validation; identified 10 interpretable predictors and quantified model transferability beyond the training distribution.

Technical Skills

Scientific ML: PyTorch, PyTorch Geometric, RDKit, SE(3)-aware modeling, GNNs, Graph Transformers, VQ-VAE, XGBoost, SHAP
LLM Systems: Context engineering, RAG, typed graph memory, multi-agent orchestration, MiniLM/MMR, context compression, vLLM, FastAPI, Docker, Pydantic/JSON Schema, LLM observability and evaluation
Programming & ML Engineering: Python, SQL, MLflow, Slurm/HPC, Git, scikit-learn, pandas, NumPy

Selected Projects

HELP-E: Long-Horizon LLM Context & Memory System [GitHub] — Mar. 2026 – Oct. 2026
Context engineering, agent memory, observability, and efficient inference — *Dallas, TX*
- Developed and containerized a FastAPI-based multi-agent LLM system integrating OpenAI-compatible vLLM model serving, typed graph memory, and MiniLM/MMR retrieval for persistent multi-session interactions.
- Built model-aware context management components supporting token-budgeted assembly, evidence prioritization, exact/semantic deduplication, and provenance-preserving memory compression.
- Instrumented system observability across request-stage latency, inference throughput, process/GPU memory utilization, context-token consumption, and LLM retries/fallbacks, with reproducible JSON audit traces for debugging and performance analysis.
- Benchmarked system quality and efficiency using paired multi-session evaluations across 2K–16K context budgets; separate graph/RAG evaluation covered 2,100 turns across 15 profiles and improved assigned-primary detection from 60% to 73%.

NLP for Web Security | HTTP Attack Detection with DistilBERT — May. 2025 – Jul. 2025
NLP for Web Security — *Dallas, TX*
- Used SQL to retrieve and filter HTTP request logs; engineered 12 security-focused features and character n-gram TF-IDF representations from 18,211 requests for four-class attack classification.
- Benchmarked logistic regression, SVM, random forest, and DistilBERT, achieving 0.84 macro F1 with DistilBERT versus 0.815 with linear SVM.

Industry Experience

Team Lead, StemiRNA Therapeutics — Apr. 2021 – Nov. 2022
Cancer Immunology – mRNA Vaccine Development — *Shanghai, China*
- Led a three-person team that designed and screened 10+ candidate mRNA vaccine sequences using six cell-based assays, advancing three leads into three in vivo antitumor-efficacy studies for KRAS-mutant lung cancer.
- Partnered with two CROs to execute three IND-enabling studies covering efficacy, toxicology, biodistribution, and dose selection, supporting a vaccine program that progressed to Phase II clinical evaluation.

Selected Publications

- **Yi Yang**, Ovidiu Daescu. "BiScale-GTR: Fragment-Aware Graph Transformers for Multi-Scale Molecular Representation Learning." *Transactions on Machine Learning Research (TMLR)*, 2026.
- **Yi Yang**, Arowa Yasmeen, Ovidiu Daescu. "Uncovering Structure–Performance Relationships in Organic Photovoltaics: Interpretable ML Model for Predicting the Power Conversion Efficiency." *IEEE Journal of Photovoltaics*, 2025.
- **Biomedical Research:** Five additional publications on the antitumor activity and mechanisms of natural compounds, including one first-author paper in a journal with impact factor >11.